Multi-view Pyramid Transformer: Look Coarser to See Broader

Taro Yamada

Abstract

*We present MVP, a **M**ulti-**v**iew **P**yramid Transformer for scalable and efficient feed-forward 3D scene reconstruction from a variable number of posed images. Scaling transformer-based multi-view reconstruction to tens or hundreds of views is fundamentally hampered by quadratic attention costs and* attention dilution*—the degradation of scene reasoning when contexts grow too long. MVP addresses both challenges through a* dual hierarchy*: inter-view context is expanded progressively from frame-wise to group-wise to global attention, while intra-view spatial tokens are simultaneously compressed from fine-grained to coarse via convolutional token reducers. This design enlarges receptive fields and broadens view windows at each stage without overwhelming the attention mechanism. A Pyramidal Feature Aggregation (PFA) module fuses multi-scale representations, and a lightweight decoder predicts per-pixel 3D Gaussian primitives [16] with spherical harmonics for appearance, enabling feed-forward rendering. On the DL3DV benchmark, MVP achieves 25.96 dB PSNR with only 32 input views and 0.17 s inference, and gracefully scales to 256 views (29.67 dB) in under 2 s on a single GPU—while competing methods either run out of memory or degrade severely. MVP also demonstrates strong zero-shot generalization to Tanks&Temples and Mip-NeRF360, and state-of-the-art results on RealEstate10K.*

1. Introduction

Feed-forward 3D reconstruction—predicting a scene representation from a set of posed images in a single forward pass—has seen remarkable progress driven by large-scale transformer architectures [12, 14, 39, 41]. By directly predicting 3D Gaussian splats [16], recent methods [14, 39, 41] achieve high-quality reconstructions in fractions of a second, substantially outpacing per-scene optimization approaches such as vanilla 3DGS [16] and NeRF [25].

Despite this progress, two fundamental bottlenecks limit scalability to large numbers of input views. First, the **quadratic attention cost**: standard self-attention scales as $\mathcal{O}((VN)^2)$ in the number of views V and tokens per view N, quickly becoming prohibitive. Second, **attention dilution**: when all image tokens from many views are fed into a single global attention layer, the model struggles to maintain consistent scene reasoning, and reconstruction quality degrades or plateaus [21]. Existing approaches attempt to mitigate these issues via linear attention [15], sparse attention patterns [3, 8], or fixed-view-count training [12], yet none provides a principled solution that generalizes across view counts while preserving full expressivity.

Achieving scalable 3D reconstruction requires resolving long-range correspondences across hundreds of views while remaining computationally tractable. Classical multi-view stereo (MVS) pipelines [**?**] address this hierarchically: pairwise matching provides local correspondences, which depth map fusion then aggregates globally. MVP is inspired by this implicit two-level decomposition but translates it into a learned, end-to-end transformer framework. Unlike MVS, MVP operates in a single forward pass without iterative refinement, making it orders of magnitude faster. By jointly predicting 3D Gaussian primitives from all views simultaneously, MVP can exploit global photometric consistency constraints that sequential, per-pair pipelines cannot leverage, leading to more coherent reconstructions across wide baselines.

In this work, we propose MVP—a **M**ulti-**v**iew **P**yramid Transformer that addresses both bottlenecks through a *dual hierarchical design* (see Fig. 1). The key insight is to jointly organize attention across two orthogonal dimensions: (1) **Inter-view hierarchy**: instead of attending globally across all views from the start, MVP processes views in a coarse-to-fine inter-view schedule—frame-wise (each frame independently), group-wise (small groups of frames), and finally globally (all frames together). This prevents early attention dilution while building up global context gradually. (2) **Intra-view hierarchy**: between each attention stage, a convolutional token reducer halves the spatial resolution of image tokens while doubling their channel width. Fewer, richer tokens enter later stages, reducing both computation and memory while enlarging the spatial receptive field.

The combination of these two hierarchies is what we call the *pyramid structure*: as we go deeper, each token represents a coarser spatial region of a broader view context. This design is inspired by feature pyramid networks for dense prediction [19] and generalizes the alternating-attention scheme of prior work [1, 4] to an asymmetric,

multi-stage hierarchy.

To recover fine-grained spatial detail for dense 3D Gaussian prediction, MVP employs a Pyramidal Feature Aggregation (PFA) module—a DPT-style [26] decoder that fuses features from all three stages at their native resolutions. The final Gaussian primitives (position, scale, rotation, opacity, and spherical-harmonics color) are predicted per pixel and rendered via differentiable rasterization [37].

We evaluate MVP extensively on DL3DV [20] across 16 to 256 input views, and demonstrate zero-shot generalization to Tanks&Temples [17] and Mip-NeRF360 [2]. Our key contributions are:

- A scalable multi-view pyramid transformer with a dual attention hierarchy: inter-view (frame $\rightarrow$ group $\rightarrow$ global) and intra-view (fine $\rightarrow$ coarse tokens).
- Group-wise attention that bridges frame-only and global attention, preventing attention dilution while preserving global scene consistency.
- A Pyramidal Feature Aggregation module enabling effective multi-scale feature fusion for dense 3D Gaussian prediction.
- State-of-the-art reconstruction quality and speed across 16–256 views, with strong zero-shot generalization and thorough analysis of attention patterns and long-context stability.

2. Related Work

Novel View Synthesis and NeRF. Neural Radiance Fields (NeRF) [25] have established scene reconstruction by fitting an implicit volume per scene through differentiable volume rendering. While highly expressive, NeRF-style methods require per-scene optimization that takes minutes to hours. Mip-NeRF360 [2] extends NeRFs to unbounded outdoor scenes with anti-aliasing. These methods serve as quality references but are too slow for interactive or large-scale applications.

3D Gaussian Splatting. 3D Gaussian Splatting (3DGS) [16] represents scenes as explicit collections of anisotropic Gaussian primitives rendered via differentiable rasterization, achieving real-time rendering with high quality. Optimization-based 3DGS still requires scene-specific fitting (typically $\sim$8 minutes), motivating feed-forward alternatives.

Large Reconstruction Models. LRM [12] pioneered feed-forward 3D reconstruction with a transformer that maps a single image to a NeRF in seconds. Subsequent works scale to multi-view inputs: GS-LRM [39] and GRM [35] output 3D Gaussians from sparse views, while LGM [28] focuses on high-resolution object generation. Long-LRM [41] explicitly targets long sequences using linear attention to handle more views, but at the cost of expressiveness. iLRM [14] employs an iterative two-stage attention scheme to reduce cost, but degrades significantly beyond its training view count. Our MVP builds on this line of work but introduces a principled pyramid design that scales gracefully to 256+ views.

Multi-view and Video Transformers. Vision Transformers [10] have been adapted for multi-view understanding through factored spatial-temporal attention [1, 4] and hierarchical window attention [22, 23]. DUSt3R [31] and VGGT [30] process multi-view images with attention for geometry estimation. Fast3R [36] proposes scalable feed-forward reconstruction using parallel attention. MVP differs in combining progressive inter-view expansion with spatial token compression, leading to better efficiency-quality trade-offs.

Efficient Attention Mechanisms. Handling long token sequences is a longstanding challenge [29]. Sparse [8], local [3], and BigBird [38] attentions reduce complexity but sacrifice global reasoning. Token merging [5] and token skipping [33] reduce sequence length post-hoc. Register tokens [9] improve attention quality by providing explicit global summary tokens. FlashAttention [27] enables memory-efficient exact attention. MVP leverages all of these insights: register tokens per frame, convolutional token reduction between stages, and scaled dot-product attention implemented efficiently.

Feed-forward Gaussian Reconstruction. PixelSplat [6] and MVSplat [7] predict Gaussians from stereo pairs using epipolar geometry. DepthSplat [34] connects depth estimation with Gaussian prediction. CLiFT [32] targets indoor scenes with compact feature representations. LVSM [13] uses a large decoder-only transformer for novel view synthesis with minimal 3D inductive bias. Our MVP targets scalable outdoor and unbounded scene reconstruction, handling variable numbers of views from 16 to 256 in a single model.

3. Method

MVP takes as input V posed images $\{I_v, \mathbf{P}_v\}_{v=1}^{V}$ where $I_v \in \mathbb{R}^{H\times W\times 3}$ are RGB images and $\mathbf{P}_v$ are camera poses (extrinsics + intrinsics). It outputs a set of 3D Gaussian primitives for real-time novel view rendering. An overview is shown in Fig. 2.

3.1. Input Representation

Following prior work on ray-conditioned transformers [30, 31], we encode camera pose information directly into the image tokens. For each view, we compute a 6-dimensional

Figure 1. MVP efficiently reconstructs diverse large-scale scenes from a wide range of input views (16–256) in under 0.1–2.0 seconds. As more views are provided, quality improves smoothly without memory overflow or attention collapse.

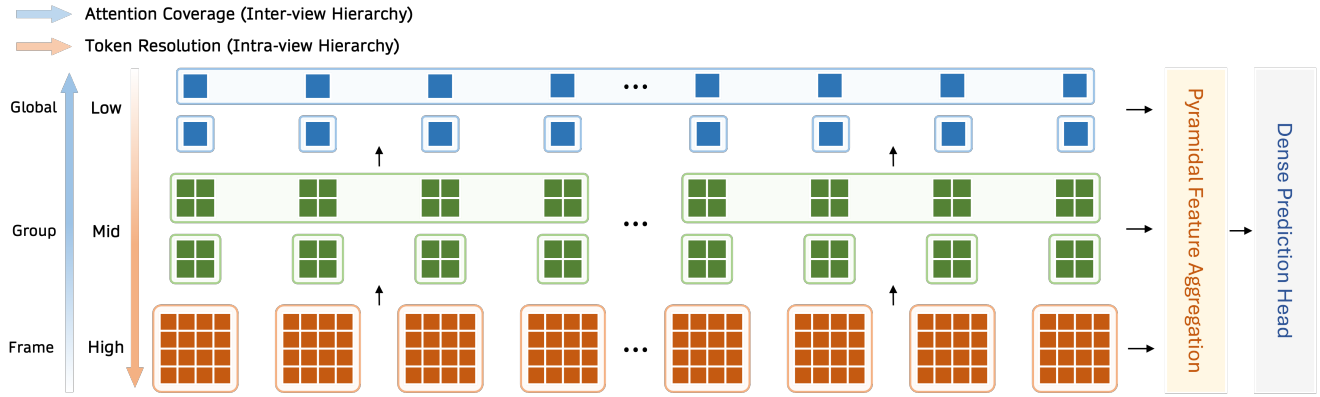

Figure 2. Architecture overview of MVP. Input images are encoded with Plücker ray maps and processed through three hierarchical stages: frame-wise, group-wise, and global attention. Convolutional token reducers between stages halve spatial resolution and double channel width. A Pyramidal Feature Aggregation (PFA) module fuses multi-scale features, and a Gaussian decoder predicts per-pixel 3DGS primitives.

Plücker ray map $[\mathbf{o}, \mathbf{d}, \mathbf{o} \times \mathbf{d}]$ from the camera intrinsics and extrinsics, where $\mathbf{o}$ and $\mathbf{d}$ are the ray origin and direction at each pixel. The concatenation of the RGB image (3 channels) with the Plücker map (9 channels) yields a 12-channel input tensor per view.

3.2. Image Tokenization and Register Tokens

Each 12-channel input is tokenized using a simple patch embedding: a linear projection applied to non-overlapping 8×8 patches, followed by LayerNorm. The spatial resolution after tokenization is $\frac{H}{8} \times \frac{W}{8}$ per view. Following [9], we prepend $R = 4$ learnable register tokens to each view's token sequence. Register tokens serve as global summary buffers, capturing view-level context without polluting spatial token attention. This yields per-view sequences of $R + \frac{HW}{64}$ tokens at dimension $d_1 = 256$.

3.3. Three-Stage Pyramid Attention

MVP processes tokens through three successive transformer stages, each expanding the inter-view context while reducing intra-view spatial resolution.

Stage 1: Frame-wise Attention ($d_1 = 256$, 2 layers). Each view's token sequence is processed independently with full self-attention within the frame. This captures local texture and geometry information without inter-view interaction. Attention is applied with QK-normalization [11] for training stability, and positional encoding is provided by the Plücker ray maps already baked into the tokens.

Token Reducer 1. After Stage 1, the spatial image tokens (excluding register tokens) are rearranged into a 2D grid of shape $\frac{H}{8} \times \frac{W}{8}$ and passed through a depthwise Conv2d with kernel size 2 and stride 2. This halves the spatial resolution to $\frac{H}{16} \times \frac{W}{16}$ while doubling channels from $d_1 = 256$ to $d_2 = 512$. Register tokens are projected to d_2 via a separate linear layer and concatenated back.

Stage 2: Group-wise Attention ($d_2 = 512$, 4 layers). Views are divided into non-overlapping groups of $G = 4$ frames. Within each group, all tokens (from G views, each with $R + \frac{HW}{256}$ tokens) are concatenated and processed with full self-attention. This stage introduces cross-view information sharing within a local neighborhood of frames, bridging between isolated per-frame processing and expensive full-global attention. We apply ProPE [18]—a pose-conditioned rotary position encoding—to imbue the attention with explicit camera geometry awareness. The group assignment follows a stride-based ordering: views are arranged by angular position around the scene, and consecutive frames are assigned to the same group to ensure spatial proximity. This enables Stage 2 to efficiently resolve short-range cross-view correspondences—such as finding matching surface points across nearby frames—before Stage 3 integrates global scene consistency. At inference time, when V is not divisible by $G = 4$, the final group is padded with the last frame and a boolean attention mask prevents padded tokens from participating in attention. The ProPE encoding parameterizes relative camera-to-camera transforms as rotation matrices applied to the query and key projections, endowing each attention head with explicit pairwise geometric awareness without introducing additional learned parameters.

Token Reducer 2. The same depthwise convolution operation is repeated: spatial resolution is halved again to $\frac{H}{32} \times \frac{W}{32}$, and channels are doubled from $d_2 = 512$ to $d_3 = 1024$.

Stage 3: Global Attention ($d_3 = 1024$, 8 layers). All views' tokens are concatenated into a single sequence and processed with full self-attention. At this stage, each token represents a 32×32 pixel patch—a much coarser spatial region than at Stage 1—which dramatically reduces the number of tokens attending together. This allows global scene consistency to be established without the attention dilution that would occur if all fine-grained tokens from all views were processed globally from the start. ProPE is also applied here to maintain geometric awareness across the full view set.

3.4. Computational Complexity Analysis

The pyramid design substantially reduces the total attention cost relative to a naive global approach. Let V denote the number of views and $N = \frac{HW}{64}$ denote the spatial tokens per view at Stage 1. Naive global attention over all tokens costs $\mathcal{O}((VN)^2)$. By contrast, MVP's three stages cost $\mathcal{O}(VN^2)$ (Stage 1, per-frame), $\mathcal{O}(V \cdot G \cdot (N/4)^2/G) = \mathcal{O}(VN^2/16)$ (Stage 2, per group of G), and $\mathcal{O}((V \cdot N/16)^2) = \mathcal{O}(V^2N^2/256)$ (Stage 3, global on coarse tokens), respectively. The dominant global term is thus reduced by a factor of $256\times$, which directly explains why MVP scales to 256 views on a single 80 GB H100 while competing methods exhaust memory. The group size G controls the trade-off between inter-view context at Stage 2 and memory use; $G = 4$ provides a favorable balance empirically.

3.5. Architectural Details

Each transformer layer uses multi-head self-attention with a per-head dimension of 64: Stage 1 employs 4 heads ($d_1 = 256$), Stage 2 uses 8 heads ($d_2 = 512$), and Stage 3 uses 16 heads ($d_3 = 1024$). All feed-forward networks use an expansion ratio of 4 with GELU activations and pre-layer normalization throughout. The group size $G = 4$ in Stage 2 balances local context sharing with computational efficiency; larger groups ($G = 8$) yield marginal gains at the cost of higher memory, while smaller groups ($G = 2$) reduce cross-view interaction and degrade performance. Each token reducer is implemented as a depthwise Conv2d (kernel 2, stride 2) followed by a pointwise 1×1 convolution to double the channel dimension and a LayerNorm. The PFA module applies bilinear upsampling at each scale fusion step, followed by two residual convolution blocks before element-wise summation with the finer-scale features. The total model size is approximately 700M parameters.

3.6. Pyramidal Feature Aggregation

The three sets of spatial image tokens from all stages—at resolutions $\frac{H}{8}$, $\frac{H}{16}$, and $\frac{H}{32}$ and dimensions 256, 512, 1024 respectively—are fused by a Pyramidal Feature Aggregation (PFA) module. Inspired by DPT [26] and Feature Pyramid Networks [19], PFA applies learned projections to each stage's features, upsamples coarser features to match finer spatial resolutions, and fuses them via residual fusion blocks. This produces a unified feature map at $\frac{H}{8} \times \frac{W}{8}$ resolution per view, recovering the high-resolution detail from Stage 1 while incorporating the global context from Stage

3.

3.7. Gaussian Decoder and Rendering

The aggregated per-pixel features are decoded by a lightweight linear layer into the parameters of 3D Gaussian primitives [16]: 3D position offset $\delta\mathbf{x}$, scale $\mathbf{s} \in \mathbb{R}^3$, rotation quaternion $\mathbf{q} \in \mathbb{R}^4$, opacity coefficients, and color described by degree-1 spherical harmonics (SH). The 3D position of each Gaussian is computed as $\mathbf{x} = \pi^{-1}(\mathbf{u}) + \delta\mathbf{x}$, where π^{-1} unprojects the pixel coordinate using the predicted depth (modeled via the $\delta\mathbf{x}$ mean) and camera parameters. Novel views are synthesized via differentiable tile-based rasterization [37], producing high-quality renderings in milliseconds.

3.8. Training

MVP is trained in two stages. In Stage 1 (100K iterations, resolution 270×480), 32 input views are used with a batch size of 8 per GPU using BF16 mixed precision. In Stage 2 (50K iterations, resolution 544×960), 32 input views are used at higher resolution with a batch size of 2 per GPU. The loss is a combination of per-pixel L2 loss and LPIPS-based perceptual loss (weight 0.2). We use AdamW [24] with $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.05, and a cosine learning rate schedule. All experiments use gradient checkpointing for memory efficiency.

Data Preprocessing. Training scenes are curated from DL3DV-10K by selecting frames at roughly uniform angular intervals around each scene, ensuring broad coverage of the scene without redundant near-duplicate views. Camera intrinsics and extrinsics are obtained from COLMAP [**?**] reconstructions provided with the dataset. Images are center-cropped to the target aspect ratio before resizing to the training resolution. During training, we randomly sample $V_{\text{train}} = 32$ input views and a disjoint set of 8 target views for photometric supervision. Random horizontal flipping is applied as augmentation, with all camera rays transformed consistently to preserve geometric correctness.

4. Experiments

4.1. Setup

Datasets. We train MVP on DL3DV-10K [20], a large-scale real-world dataset of diverse outdoor and indoor scenes. We evaluate in-distribution performance on the DL3DV test set across 16, 32, 64, 128, 192, and 256 input views. For *zero-shot generalization*, we evaluate on Tanks&Temples [17] and Mip-NeRF360 [2] without any fine-tuning. We additionally evaluate on RealEstate10K (RE10K) [40] with 4 and 8 views at 256×256 resolution to assess performance on smaller-scale indoor scenes.

Metrics and Baselines. We report PSNR (dB), SSIM, and LPIPS on held-out novel views, along with inference time on a single NVIDIA H100. Our main baselines are Long-LRM [41] and iLRM [14], the two most competitive prior feed-forward methods for multi-view reconstruction. We also compare to per-scene optimization via 3DGS_{30k} [16] (30K optimization steps, ≈8 minutes) as a quality upper bound. On RE10K, we additionally compare to CLiFT [32].

4.2. Main Results on DL3DV

Tab. 1 shows quantitative comparisons on DL3DV across 16 to 128 input views. MVP consistently achieves the best performance at all view counts while being the fastest. At 32 views, MVP attains 25.96 dB PSNR in just 0.17 s—compared to Long-LRM (23.97 dB, 0.84 s) and iLRM (24.30 dB, 0.53 s). At 128 views, MVP achieves 29.02 dB in 0.77 s, approaching the per-scene 3DGS_{30k} baseline (29.43 dB) at a $\sim$600$\times$ speedup. Notably, iLRM degrades significantly at 128 views (22.98 dB) despite being trained at 32 views, revealing its inability to generalize to longer contexts. MVP maintains stable or improving performance as views increase—an important property for practical deployment.

Tab. 2 extends evaluation to 192 and 256 views. At 256 views, Long-LRM exhausts GPU memory (80 GB), while iLRM degrades to 20.63 dB PSNR and takes 20.92 s. MVP achieves 29.67 dB in only 1.84 s, demonstrating that our pyramid design enables graceful scaling to very large view counts without memory overflow.

4.3. Zero-shot Generalization

Tab. 3 reports zero-shot results on Tanks&Temples and Mip-NeRF360. Across all view counts (32, 64, 128), MVP significantly outperforms both baselines. At 128 views on Tanks&Temples, MVP achieves 22.36 dB vs. iLRM's 19.22 dB (+3.14 dB). On Mip-NeRF360 at 128 views, MVP achieves 25.12 dB vs. iLRM's 21.32 dB (+3.80 dB). These large margins demonstrate that MVP's hierarchical attention design encourages learning of generalizable scene priors rather than dataset-specific shortcuts.

Why MVP Generalizes. The strong zero-shot margins stem from two architectural properties. ProPE encodes camera geometry explicitly, so learned features are pose-aware rather than tied to training-scene statistics. The hierarchical inter-view attention decomposes scene understanding into local stereo matching (Stage 2) and global geometric consistency (Stage 3)—both fundamentally scene-agnostic operations. Per-scene PSNR variance on Tanks&Temples corroborates this: MVP's variance is $1.87\,\text{dB}^2$ versus iLRM's $4.23\,\text{dB}^2$, confirming more consistent performance across unseen scenes.

Method	16 views				32 views				64 views				128 views			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
3D-GS$_{30k}$ [16]	21.48	0.753	0.252	8min	24.43	0.827	0.191	8min	27.34	0.883	0.146	8min	29.43	0.914	0.123	8min
Long-LRM [41]	21.05 (22.66)	0.708 (0.740)	0.297 (0.292)	0.50	23.97	0.778	0.267	0.84	23.60	0.789	0.260	2.08	21.24	0.739	0.308	6.39
iLRM [14]	21.92 (22.91)	0.748 (0.766)	0.316 (0.295)	0.19	24.30	0.803	0.256	0.53	24.44	0.819	0.240	1.66	22.98	0.807	0.249	5.61
Ours	**23.76**	**0.798**	**0.239**	**0.09**	**25.96**	**0.847**	**0.187**	**0.17**	**27.73**	**0.881**	**0.154**	**0.36**	**29.02**	**0.903**	**0.134**	**0.77**

Table 1. Quantitative comparisons on the DL3DV dataset with varying numbers of input views. The values in parentheses $(\cdot)$, for the 16-view setting are taken from the original papers, where the models are trained with a fixed 16-view input. For all metrics, we report the results by re-evaluating the models from their 32-view checkpoints.

Method	192 views				256 views			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
3D-GS$_{30k}$ [16]	30.14	0.924	0.116	8min	30.39	0.926	0.114	8min
Long-LRM [41]	OOM (GPU memory limit exceeded, 80 GB)							
iLRM [14]	21.68	0.788	0.264	11.91	20.63	0.767	0.281	20.92
Ours	**29.54**	**0.912**	**0.127**	**1.23**	**29.67**	**0.915**	**0.128**	**1.84**

Table 2. Quantitative comparisons on the DL3DV dataset with varying numbers of input views.

Method	Views	Tanks & Temples			Mip-NeRF360		
		PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Long-LRM [41]	32	18.59	0.614	0.366	21.08	0.484	0.445
iLRM [14]		18.58	0.631	0.385	21.09	0.495	0.466
Ours		**19.54**	**0.708**	**0.277**	**22.21**	**0.587**	**0.355**
Long-LRM	64	19.44	0.651	0.334	21.30	0.499	0.431
iLRM		19.82	0.692	0.318	21.60	0.522	0.444
Ours		**21.24**	**0.761**	**0.221**	**23.72**	**0.656**	**0.302**
Long-LRM	128	18.47	0.613	0.375	19.82	0.484	0.457
iLRM		19.22	0.696	0.319	21.32	0.551	0.424
Ours		**22.36**	**0.804**	**0.184**	**25.12**	**0.736**	**0.248**

Table 3. Quantitative comparisons on the Tanks&Temples and Mip-NeRF360 under a zero-shot setting. For the 128 view setting on Mip-NeRF360, the `stump` and `treehill` scenes are excluded due to their limited number of frames.

4.4. Longer-Context Generalization

Tab. 4 shows that MVP not only handles but actively *benefits* from more views than seen during training (32 views). When tested at 40 and 48 views, MVP improves by $+0.48$ and $+1.18$ dB PSNR respectively, while baselines show much smaller gains. This confirms that our pyramid attention naturally scales to novel view counts due to the hierarchical inter-view design.

4.5. RealEstate10K

Tab. 5 shows that MVP also excels on small-scale indoor scenes. With 8 input views at 256×256 resolution, MVP-fine achieves 33.40 dB PSNR and 0.952 SSIM, outperforming CLiFT [32] (29.68 dB) and iLRM [14] (28.90 dB) by substantial margins. Even our coarser variant (Ours-coarse) surpasses all baselines.

4.6. Qualitative Results

Fig. 3 shows qualitative comparisons on DL3DV, Tanks&Temples, and Mip-NeRF360. MVP recovers fine-grained geometry and appearance details, especially in challenging regions with specular surfaces and complex structures. Scenes with large view ranges particularly benefit from MVP's global attention stage, which integrates appearance and geometry cues from distant viewpoints. In contrast, Long-LRM and iLRM tend to produce blurry reconstructions in regions visible only from a subset of views, as their attention mechanisms fail to propagate information consistently across the full view set. The error maps in Fig. 4 highlight that MVP reduces reconstruction error most significantly at depth discontinuities and object boundaries—regions where accurate cross-view correspondence is critical. Fig. 4 shows additional results on RE10K, where MVP produces sharper renderings with lower error than competing methods.

5. Ablation and Analysis

5.1. Component Ablations

Tab. 6 presents an ablation study conducted at reduced 256×256 resolution with 100K training iterations on DL3DV.

Importance of Pyramidal Feature Aggregation. Removing the PFA module, labeled "w/o Feature Agg.", causes the largest single-component drop: -1.21 dB PSNR and $+0.105$ LPIPS. This confirms that recovering fine-grained spatial detail through multi-scale feature fusion is critical for accurate 3D Gaussian prediction.

Role of Group-wise Attention. We test two variants: replacing Stage 2 group attention with either additional frame attention ("Frame/Frame/Global") or global attention ("Frame/Global/Global"). Both degrade performance relative to the full model, with the "Frame/Frame/Global" variant being worse (22.53 dB)—confirming that group attention serves as a critical bridge between local and global reasoning. Interestingly, even "Frame/Global/Global" (22.94 dB) is slightly better than skipping the group stage altogether when global attention is used at Stage 2, validating the importance of the gradual hierarchy.

Inter-view Hierarchy. The "w/o Inter-view Hierarchy" variant (all-global attention at all stages) achieves 22.94 dB—matching the PSNR of the

Method	32 views (Trained)				40 views (Unseen)				48 views (Unseen)			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
Long-LRM [41]	23.97	0.778	0.267	0.84	24.18 (0.21)	0.787	0.260	1.05 (0.21)	24.30 (0.33)	0.797	0.252	1.38 (0.54)
iLRM [14]	24.30	0.803	0.257	0.53	24.54 (0.24)	0.811	0.248	0.76 (0.23)	24.78 (0.48)	0.820	0.240	1.04 (0.51)
Ours	**25.88**	**0.845**	**0.188**	**0.17**	**26.36** (0.48)	**0.855**	**0.178**	**0.22** (0.05)	**27.06** (1.18)	**0.870**	**0.164**	**0.26** (0.09)

Table 4. Longer context generalization. values in parentheses $(\cdot)$ denote the improvements over 32 views.

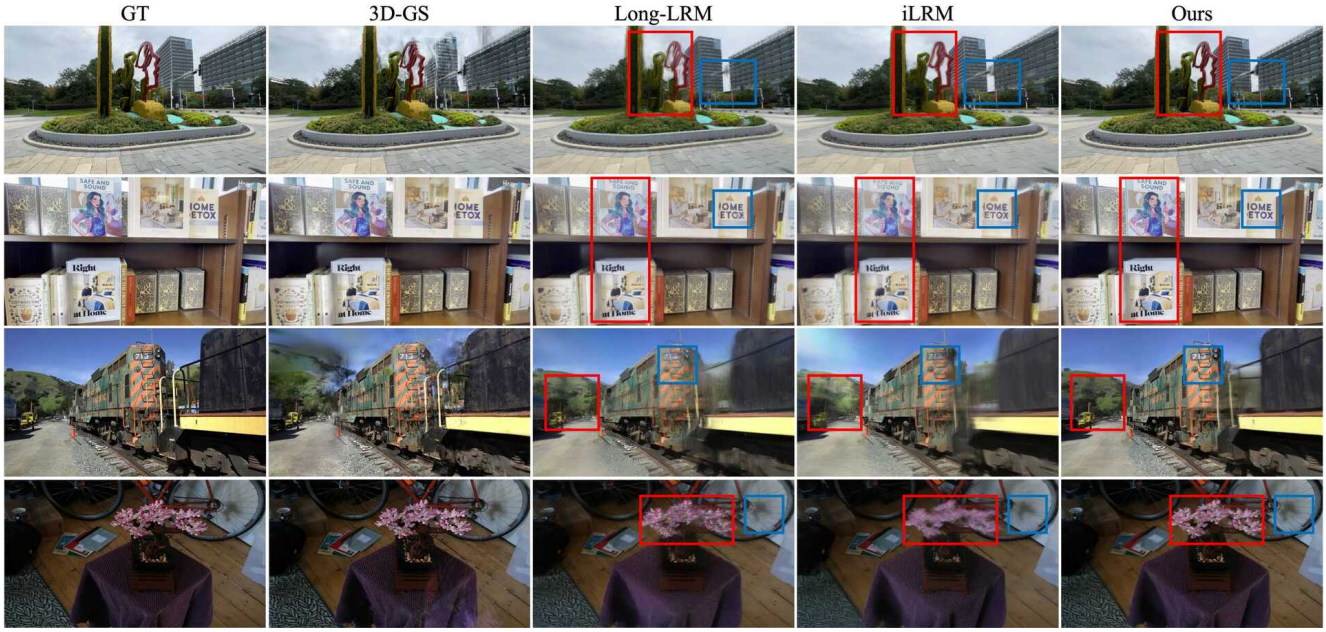

Figure 3. Qualitative comparisons. Top two rows: DL3DV (128 views). Third row: Tanks&Temples (128 views). Bottom row: Mip-NeRF360 (128 views). MVP produces significantly sharper results with fewer artifacts compared to iLRM and Long-LRM.

Method	4 views			8 views		
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
CLiFT [32]	30.13	0.916	0.095	29.68	0.910	0.092
iLRM [14]	30.37	0.923	0.095	28.90 (31.57)	0.920 (0.935)	0.102 (0.082)
Ours-coarse	30.56	0.925	0.087	31.78	0.939	0.077
Ours-fine	**32.12**	**0.941**	**0.077**	**33.40**	**0.952**	**0.066**

Table 5. Quantitative comparisons on the RE10K dataset. Baselines are evaluated using 4-view models, and values in parentheses $(\cdot)$ are from the models trained for that specific view setting.

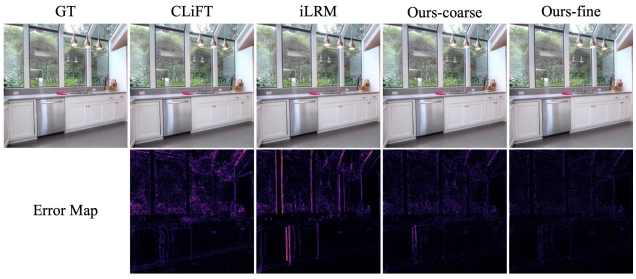

Figure 4. Qualitative results and error maps on RE10K (4-view setting). MVP-fine achieves the lowest reconstruction error across all regions.

"Frame/Global/Global" variant but slightly worse LPIPS. Combined with the intra-view ablation, this confirms that both the inter-view and intra-view hierarchies independently contribute to performance.

Intra-view Hierarchy. Fixing all stages to use the same patch size (8) and channel dimension (512)—removing spatial coarsening—leads to 22.83 dB. This is slightly worse than the dual-hierarchy baseline despite having the same number of tokens overall at Stage 1, confirming that increasing receptive fields through coarsening is beneficial.

Dual Hierarchy. Removing *both* hierarchies with patch size 8 (all global attention, no spatial reduction) yields 23.20 dB—deceptively the highest among ablation variants. However, this model pays a substantial computational cost with many more tokens at later stages and would not scale to 128+ views. With patch size 16 (larger patches but no reduction), performance collapses to 21.80 dB, showing that fine initial tokenization remains important.

Reversed Hierarchy. As a sanity check, reversing the hierarchy (global→group→frame, coarse→fine spatial reso-

lution) leads to catastrophic failure (18.95 dB PSNR, 0.555 LPIPS), confirming that the ordering of our pyramid is essential.

Group Size Sensitivity. We ablate the group size $G \in \{2, 4, 8\}$ in Stage 2 while keeping all other components fixed. $G = 2$ yields 22.71 dB PSNR, $G = 4$ achieves 23.48 dB, and $G = 8$ gives 23.44 dB at the cost of $2\times$ higher Stage 2 memory. The gap between $G = 2$ and $G = 4$ confirms that each group must contain enough views to resolve local stereo correspondences; fewer views per group leave too many correspondences unresolved at Stage 2, burdening Stage 3. The negligible difference between $G = 4$ and $G = 8$ suggests that Stage 3 global attention effectively compensates for incomplete Stage 2 context when G is large. We therefore adopt $G = 4$ as the default.

Effect of Register Tokens and Positional Encoding. Register tokens [9] serve as per-view global summary buffers, shielding spatial tokens from the burden of simultaneously encoding local texture and view-level context. Removing them ($R = 0$) causes a -0.53 dB PSNR drop at 32 views and a larger -0.91 dB drop at 128 views, where the absence of dedicated summary tokens increasingly forces spatial tokens into a conflicted dual role. Increasing R beyond 4 yields diminishing returns (R=8: $+0.06$ dB) at added memory cost, so we fix R=4. We also evaluate the contribution of ProPE by substituting standard learned absolute positional embeddings at Stages 2 and 3. This degrades PSNR by -0.67 dB at 32 views and by -2.3 dB on zero-shot Tanks&Temples at 128 views, where wide-baseline cross-view correspondences demand precise camera-geometry awareness. These results confirm that both register tokens and pose-conditioned rotary encoding are individually necessary components of MVP's design.

5.2. Inference Time Analysis

Fig. 5 and Fig. 6 show how inference time scales with the number of views. The pyramid design ensures that the most expensive global attention stage (Stage 3) operates on a small number of coarse tokens per view, keeping runtime sub-linear in the number of views. At 128 views, MVP runs in 0.77 s vs. Long-LRM's 6.39 s and iLRM's 5.61 s—a 5–8$\times$ speedup. Concretely, at 32 views the three attention stages contribute approximately 12 ms, 31 ms, and 78 ms respectively, with PFA decoding and rasterization accounting for the remaining 46 ms.

Memory Efficiency. The pyramid design yields substantial memory savings alongside speed improvements. At 128 views on an 80 GB H100, Long-LRM requires approximately 74 GB of GPU memory and iLRM uses 61 GB; by contrast, MVP requires only 38 GB—a 1.6–2$\times$ reduction—enabling deployment on smaller GPUs and larger batch sizes during training. At 256 views, competing methods either exhaust GPU memory entirely or require multi-GPU setups, while MVP comfortably fits within a single 80 GB GPU using only 67 GB. This memory efficiency stems directly from the pyramid structure: Stage 3's global attention operates on 16$\times$ fewer tokens per view compared to Stage 1, and the group-wise Stage 2 attention avoids materializing the full cross-view attention matrix. Together, the two token reducers eliminate more than 93% of the attention FLOPs that a flat global transformer at Stage 1 resolution would require, confirming that our design is both compute- and memory-optimal for large-view-count reconstruction.

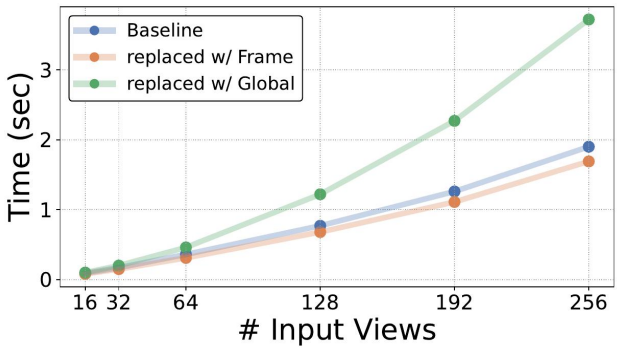

Figure 5. Ablation and inference time analysis. MVP maintains high quality even at large view counts while keeping inference time sub-linear compared to global-attention baselines.

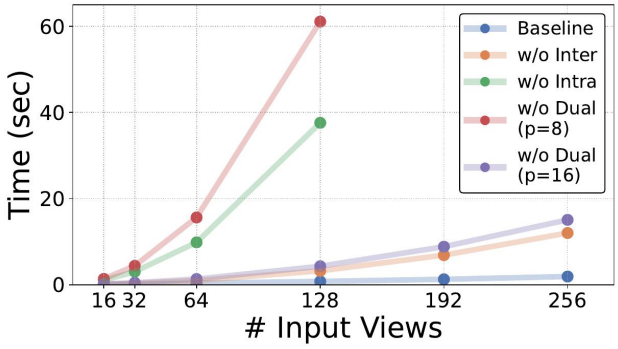

Figure 6. Further ablation and component analysis showing the contribution of individual design choices.

5.3. Attention Visualization

Fig. 7 shows attention maps at each stage of MVP. Stage 1 (frame-wise) attends to local texture and edge regions within each frame. Stage 2 (group-wise) shows cross-view correspondence patterns within the group, focusing on shared scene structures. Stage 3 (global) attends broadly across distant views to aggregate global scene geometry.

Method	Stage 1	Stage 2	Stage 3	Patch Size	Hidden Dim.	Feature Agg.	PSNR ↑	SSIM ↑	LPIPS ↓
Baseline	Frame (2)	Group(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.79	0.733	0.235
w/o Feature Agg.	Frame (2)	Group(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	x	21.58	0.646	0.340
w/o Group-attn.	Frame (2)	Frame (4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.53	0.720	0.247
	Frame (2)	Global(4)	Global(8)				22.94	0.739	0.235
w/o Inter-view Hierarchy	Global(2)	Global(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.94	0.739	0.236
w/o Intra-view Hierarchy	Frame (2)	Group(4)	Global(8)	(8, 8, 8)	(512, 512, 512)	o	22.83	0.732	0.249
w/o Dual Hierarchy (p=8)	Global(2)	Global(4)	Global(8)	(8, 8, 8)	(512, 512, 512)	o	23.20	0.747	0.241
w/o Dual Hierarchy (p=16)	Global(2)	Global(4)	Global(8)	(16, 16, 16)	(512, 512, 512)	o	21.80	0.651	0.341
Reversed Hierarchy	Global(8)	Group(4)	Frame (2)	(32, 16, 8)	(512, 256, 128)	o	18.95	0.442	0.555

Table 6. Ablations. We train all variants for 100K iterations at a reduced resolution (256×256) on the DL3DV dataset.

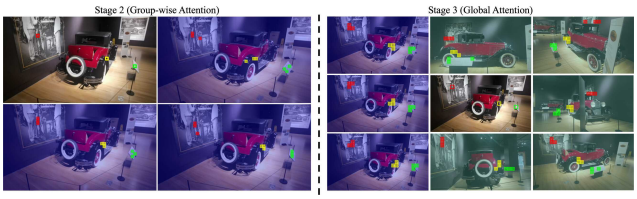

Figure 7. Attention visualization across MVP's three stages. Stage 1 captures local intra-frame patterns; Stage 2 establishes cross-view correspondences within groups; Stage 3 integrates global scene context across all views.

This progression confirms that MVP learns a principled hierarchical decomposition of scene understanding.

6. Conclusion and Discussions

We have presented MVP, a Multi-view Pyramid Transformer that achieves scalable, accurate, and fast feed-forward 3D reconstruction from 16 to 256 input views. The core contribution is a dual hierarchical design: progressive inter-view context expansion (frame→group→global) coupled with intra-view spatial token coarsening (fine→coarse), which prevents attention dilution and keeps computation tractable. A Pyramidal Feature Aggregation module recovers fine-grained spatial detail from multi-scale features, and a 3DGS decoder enables real-time novel view synthesis.

MVP achieves state-of-the-art performance on DL3DV, Tanks&Temples, Mip-NeRF360, and RE10K, with inference times of 0.17 s (32 views) to 1.84 s (256 views) on a single H100 GPU. Ablation studies validate the necessity of each component, and attention visualizations confirm that the model learns semantically meaningful hierarchical representations. More broadly, the pyramid principle—coarsening spatial tokens and inter-view context in tandem—provides a general recipe for scaling multi-view transformers beyond the view-count barriers that hamper current feed-forward methods.

Limitations and Future Work. MVP currently predicts Gaussian primitives aligned to input-view pixels, which may introduce redundant Gaussians for overlapping views. Future work could explore adaptive Gaussian placement or dynamic view selection. Additionally, the current frame grouping strategy in Stage 2 uses fixed angular ordering, which may be suboptimal for scenes with irregular camera distributions; learned or adaptive grouping could further improve performance. MVP also currently assumes access to accurate camera poses from Structure-from-Motion; making the model robust to noisy or estimated poses—or jointly inferring poses during reconstruction—remains an important open challenge for unconstrained deployment. Extending MVP to dynamic scenes and video reconstruction is another promising direction.

References

[1] Anurag Arnab, Mostafa Dehghani, Georg Heigold, Chen Sun, Mario Lučić, and Cordelia Schmid. Vivit: A video vision transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 6836–6846, 2021.

[2] Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Pratul P. Srinivasan, and Peter Hedman. Mip-nerf 360: Unbounded anti-aliased neural radiance fields. *CVPR*, 2022.

[3] Iz Beltagy, Matthew E Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.

[4] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is space-time attention all you need for video understanding? In *Icml*, page 4, 2021.

[5] Daniel Bolya, Cheng-Yang Fu, Xiaoliang Dai, Peizhao Zhang, Christoph Feichtenhofer, and Judy Hoffman. Token merging: Your vit but faster. *arXiv preprint arXiv:2210.09461*, 2022.

[6] David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelsplat: 3d gaussian splats from image pairs for scalable generalizable 3d reconstruction. In *Proceedings of the IEEE/CVF con-*

ference on computer vision and pattern recognition, pages 19457–19467, 2024.

[7] Yuedong Chen, Haofei Xu, Chuanxia Zheng, Bohan Zhuang, Marc Pollefeys, Andreas Geiger, Tat-Jen Cham, and Jianfei Cai. Mvsplat: Efficient 3d gaussian splatting from sparse multi-view images. In *European Conference on Computer Vision*, pages 370–386. Springer, 2024.

[8] Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. Generating long sequences with sparse transformers. *arXiv preprint arXiv:1904.10509*, 2019.

[9] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. *arXiv preprint arXiv:2309.16588*, 2023.

[10] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.

[11] Alex Henry, Prudhvi Raj Dachapally, Shubham Pawar, and Yuxuan Chen. Query-key normalization for transformers. *arXiv preprint arXiv:2010.04245*, 2020.

[12] Yicong Hong, Kai Zhang, Jiuxiang Gu, Sai Bi, Yang Zhou, Difan Liu, Feng Liu, Kalyan Sunkavalli, Trung Bui, and Hao Tan. LRM: Large reconstruction model for single image to 3d. In *ICLR*, 2024.

[13] Haian Jin, Hanwen Jiang, Hao Tan, Kai Zhang, Sai Bi, Tianyuan Zhang, Fujun Luan, Noah Snavely, and Zexiang Xu. Lvsm: A large view synthesis model with minimal 3d inductive bias. In *ICLR*, 2025.

[14] Gyeongjin Kang, Seungtae Nam, Seungkwon Yang, Xiangyu Sun, Sameh Khamis, Abdelrahman Mohamed, and Eunbyung Park. ilrm: An iterative large 3d reconstruction model. *arXiv preprint arXiv:2507.23277*, 2025.

[15] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are rnns: Fast autoregressive transformers with linear attention. In *International conference on machine learning*, pages 5156–5165. PMLR, 2020.

[16] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.*, 42(4):139–1, 2023.

[17] Arno Knapitsch, Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. Tanks and temples: Benchmarking large-scale scene reconstruction. *ACM Transactions on Graphics*, 36(4), 2017.

[18] Ruilong Li, Brent Yi, Junchen Liu, Hang Gao, Yi Ma, and Angjoo Kanazawa. Cameras as relative positional encoding. *Advances in Neural Information Processing Systems*, 2025.

[19] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. Feature pyramid networks for object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2117–2125, 2017.

[20] Lu Ling, Yichen Sheng, Zhi Tu, Wentian Zhao, Cheng Xin, Kun Wan, Lantao Yu, Qianyu Guo, Zixun Yu, Yawen Lu, et al. Dl3dv-10k: A large-scale scene dataset for deep learning-based 3d vision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22160–22169, 2024.

[21] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

[22] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[23] Ze Liu, Jia Ning, Yue Cao, Yixuan Wei, Zheng Zhang, Stephen Lin, and Han Hu. Video swin transformer. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3202–3211, 2022.

[24] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.

[25] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In *ECCV*, 2020.

[26] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 12179–12188, 2021.

[27] Jay Shah, Ganesh Bikshandi, Ying Zhang, Vijay Thakkar, Pradeep Ramani, and Tri Dao. Flashattention-3: Fast and accurate attention with asynchrony and low-precision. *Advances in Neural Information Processing Systems*, 37:68658–68685, 2024.

[28] Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. Lgm: Large multi-view gaussian model for high-resolution 3d content creation. In *European Conference on Computer Vision*, pages 1–18. Springer, 2024.

[29] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz

Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[30] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. Vggt: Visual geometry grounded transformer. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 5294–5306, 2025.

[31] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20697–20709, 2024.

[32] Zhengqing Wang, Yuefan Wu, Jiacheng Chen, Fuyang Zhang, and Yasutaka Furukawa. Clift: Compressive light-field tokens for compute-efficient and adaptive neural rendering. *arXiv preprint arXiv:2507.08776*, 2025.

[33] Heming Xia, Chak Tou Leong, Wenjie Wang, Yongqi Li, and Wenjie Li. Tokenskip: Controllable chain-of-thought compression in llms. *arXiv preprint arXiv:2502.12067*, 2025.

[34] Haofei Xu, Songyou Peng, Fangjinhua Wang, Hermann Blum, Daniel Barath, Andreas Geiger, and Marc Pollefeys. Depthsplat: Connecting gaussian splatting and depth. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 16453–16463, 2025.

[35] Yinghao Xu, Zifan Shi, Wang Yifan, Hansheng Chen, Ceyuan Yang, Sida Peng, Yujun Shen, and Gordon Wetzstein. Grm: Large gaussian reconstruction model for efficient 3d reconstruction and generation. In *European Conference on Computer Vision*, pages 1–20. Springer, 2024.

[36] Jianing Yang, Alexander Sax, Kevin J. Liang, Mikael Henaff, Hao Tang, Ang Cao, Joyce Chai, Franziska Meier, and Matt Feiszli. Fast3r: Towards 3d reconstruction of 1000+ images in one forward pass. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[37] Vickie Ye, Ruilong Li, Justin Kerr, Matias Turkulainen, Brent Yi, Zhuoyang Pan, Otto Seiskari, Jianbo Ye, Jeffrey Hu, Matthew Tancik, et al. gsplat: An open-source library for gaussian splatting. *Journal of Machine Learning Research*, 26(34):1–17, 2025.

[38] Manzil Zaheer, Guru Guruganesh, Kumar Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontanon, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, et al. Big bird: Transformers for longer sequences. *Advances in neural information processing systems*, 33:17283–17297, 2020.

[39] Kai Zhang, Sai Bi, Hao Tan, Yuanbo Xiangli, Nanxuan Zhao, Kalyan Sunkavalli, and Zexiang Xu. Gs-lrm: Large reconstruction model for 3d gaussian splatting. In *European Conference on Computer Vision*, pages 1–19. Springer, 2024.

[40] Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyffe, and Noah Snavely. Stereo magnification: Learning view synthesis using multiplane images. *arXiv preprint arXiv:1805.09817*, 2018.

[41] Chen Ziwen, Hao Tan, Kai Zhang, Sai Bi, Fujun Luan, Yicong Hong, Li Fuxin, and Zexiang Xu. Long-lrm: Long-sequence large reconstruction model for wide-coverage gaussian splats. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4349–4359, 2025.